REAL WORLD EXAMPLE: THE NOKIA NETWORK DIGITAL TWIN

Nokia’s Network Digital Twin takes data about the performance of a communications network, including latency, throughput, signal strength and signal to noise ratio from telemetry sent directly from devices on the network and performs threshold monitoring on this representation of the state of the network against key performance indicators (KPIs). In the case of a threshold being breached, such as a change in topography reducing network coverage, network planners are alerted so that they can perform actions to ensure network connectivity is maintained.

1.4.4 Supporting the full infrastructure asset lifecycle

The preceding advantages coalesce in a single, evolving digital twin that supports physical infrastructure throughout its entire lifecycle. The twin starts as a planning and design tool, allowing engineers to simulate different scenarios, test performance, and optimize configurations before any physical work begins. As the asset moves into the construction phase, the twin transitions into a verification and tracking system, recording progress, validating installations against initial designs, and capturing "as-built" information. Finally, once the asset is operational, the digital twin becomes a real-time monitoring and optimization platform, continuously ingesting live sensor data to track performance, predict future maintenance needs, and enable adaptive, data-driven management for the rest of its service life.

REAL WORLD EXAMPLE: CROSS RIVER RAIL PROJECT

The cross river rail project in Brisbane Australia, was a multi billion dollar infrastructure project designed to remove bottlenecks in the existing public transport system. The project mandated a common data environment underpinning a digital twin, to which all contractors were obliged to contribute right from the planning stage, through construction, to operation. The benefits of the resulting digital twin included early identification of design issues, reconciliation across multiple major works packages, the ability to brief stakeholders effectively, and overcoming environmental issues by being able to travel through as-yet built environments, first-person and at scale.